



CO2 AT2 ASSESSMENT.docx



File View Tools Help

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Page 1 / 9 100%

1) Given

$$A = +70 \Rightarrow 01000110_2$$

$$B = +80 \Rightarrow 01010000_2$$

Step 1: Add the binary numbers:

$$\begin{array}{r} 01000110 \\ + 01010000 \\ \hline 10010110 \end{array}$$

Result = 10010110_2

Step 2: Find Decimal Sum

$$A = 70$$

$$B = 80$$

$$\text{Sum} = 70 + 80$$

$$= 150$$

Step 3: Check 8-bit Range:

$$-128 \text{ to } +127$$

obtained Sum = 150

$$150 > 127$$

Therefore, overflow occurs

Step 4: Check Sign bits

$$A = 01000110 \rightarrow \text{Sign bit: } 0$$

$$B = 01010000 \rightarrow \text{Sign bit: } 0$$

$$\text{Result} = 10010110 \rightarrow \text{Sign bit: } 1$$

Two positive numbers produce a negative result.